

DR6g: The Fully Domain-Blind Verifier Also Escaped – And Refines DR5 Again

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – DR6g **Status:** Overall **NO_GO** on the "further degradation" gate suite. R6 actually scored *higher* (10 vs DR6f's 8). Realisation-placebo separation preserved. Prediction refuted – LLM verifier escapes the wall on a *domain-general* signal (implicit-vs-explicit code structure), not on D-specific semantic reasoning. **Date:** 2026-07-27

Abstract

DR6f showed partial wall degradation when \$D\$ was withheld from the prompt but domain framing was kept. Extrapolating to a fully domain-blind prompt, I predicted DR6g would see the wall substantially return – the semantic-access escape route requires knowledge of \$D\$'s domain, or so DR7's independent-semantic-access reasoning suggested.

The prediction was wrong. With a prompt that only asked *"how likely does this code make any implicit assumption about how the world works that isn't explicit in its imports, types, or comments,"* Claude verifiers scored realisations at 8–10 (median 9.0) and placebos at 0–2 (median 2, max 2). Overlap gap = 6. R6 scored **10** (higher than DR6f's 8). Wall did not reappear.

What Claude was doing: picking up a domain-general signal – implicit-vs-explicit code structure – that correlates with the naive-UTC target commitment because the target commitment IS an implicit assumption. Code that uses `pytz.utc.localize()` reveals its assumption explicitly. Code that uses `datetime.utcnow()` or `datetime(year=..., ...)` hides its assumption. The LLM can distinguish these without knowing that the specific assumption is about UTC.

The refined-refined DR5 claim: semantic access to \$D\$ (DR5 condition (b)) can be *proxied* by any signal that correlates with \$D\$'s surface structure – including domain-general signals like "this code is implicit rather than explicit." For naive-UTC, the implicit-vs-explicit axis correlates strongly enough with \$D\$ that the verifier escapes the wall via the proxy alone.

Consequence. The wall's bite depends not just on semantic access to \$D\$ specifically, but on any accessible signal that discriminates \$D\$-realisations from non-realisations. For code correctness targets that align with implicit-vs-explicit structure – a large class – even domain-blind LLM verifiers escape. For code correctness targets that *don't* align with any domain-general signal (say, "this code has a subtle deadlock" – no reliable implicit/explicit correlate), the wall should still bite.

1. Setup

Same 11 snippets as DR6d/DR6e/DR6f. Three sandboxed Claude subagents. Prompt strips all domain-specific framing:

For each Python code snippet, score how likely it is that the code makes ANY implicit assumption about how the world works – a fact about the runtime environment, external systems, data conventions, or user context – that is not explicit in its imports, types, or comments.

No mention of `datetime`, `timezone`, `UTC`, or naive-UTC. Domain-blind.

2. Results

Per-snippet consensus scores across three verifiers:

snippet	kind	DR6e (D given)	DR6f (D-adjacent)	DR6g (blind)
R1_utcnow_direct	realisation	10	10	9
R2_replace_tzinfo_none	realisation	9	10	9
R3_time_fromtimestamp	realisation	8	9	9
R4_iso_parse_no_tz	realisation	8	9	8
R5_combine_utc_convention	realisation	10	8	9
R6_json_field_no_tz	realisation	10	8	10
P1_pytz_localize	placebo	1	2	2
P2_now_with_timezone_utc	placebo	0	1	1
P3_zoneinfo_user_tz	placebo	0	1	2
P4_no_datetime_arithmetic	placebo	0	0	0
P5_arrow_aware	placebo	0	1	2

Aggregate:

- DR6g realisations: median 9.0, min 8, max 10.
- DR6g placebos: median 2, max 2.
- Overlap gap: $8 - 2 = 6$ (same as DR6f).

3. Gate decisions

gate		
G1 median below DR6f	NO_GO	$9.0 = 9.0$
G2 gap below DR6f	NO_GO	$6 = 6$
G3 R6 below DR6f	NO_GO	$10 > 8$
G4 wall present direct	NO_GO	overlap gap +6

Overall NO_GO. Licensed reading:

domain_blind_still_discriminates: Claude preserved discrimination even without domain framing. LLM verifier is doing more than semantic reasoning about D – possibly recognising implicit-vs-explicit style as a domain-independent signal.

4. What DR6g refines about DR5

DR7's independent-semantic-access requirement can be satisfied by **proxy signals** that don't specifically encode $\$D\'s semantic content.

For naive-UTC (the DR6 target), the proxy is *implicit-vs-explicit code structure*. Realisations of naive-UTC are, by definition, code that does not carry explicit timezone information. Placebos are code that does. Any verifier that can distinguish implicit from explicit code – which any competent LLM can, and which is a domain-general skill – escapes the wall on the naive-UTC target *even without knowing what naive-UTC is*.

The re-refined-refined DR5 (call this DR5):*

The wall bites when

>

(a) the realisation set is open-ended at test time, AND

(b) the verifier has no signal – semantic reasoning about $\$D\'s , or

domain-general proxy correlated with \$D\$'s realisation structure
– that discriminates realisations from non-realisations.

Condition (b) is now weaker than "verifier reasons semantically about \$D\$." It is "verifier has any signal that separates realisations from non-realisations, whether or not it knows \$D\$."

This weakening does not vacate the theorem. Many \$D\$'s have no domain-general proxy – DCR1f's T1 (absolute simultaneity in 1900 physics) is exactly such a \$D\$, because implicit-vs-explicit is not a distinction that separates T1-realizations from T2-realizations or from non-realizations. The wall bites on T1 precisely because no proxy signal works.

Consequence: the wall's applicability depends on the \$D\$–proxy correlation structure of the target commitment, not just on whether the verifier "knows \$D\$." DR5*'s condition (b) collapses to DR5's condition (b) when no proxy exists; it is weaker when one does.

5. The updated gradient

Across DR6e/f/g, plotting R6 score against prompt specificity:

prompt	R6 score	overlap gap
D-specified (DR6e)	10	7
D-adjacent domain (DR6f)	8	6
Fully domain-blind (DR6g)	10	6

The gradient is **non-monotonic**. DR6f dips below DR6g because the D-adjacent prompt is worse than either extreme: it primes datetime attention without giving the specific commitment, so Claude might reach for surface features that don't quite capture \$D\$ (e.g., "does this use datetime?" – which fires on both realizations and placebos). The domain-blind prompt is cleaner because Claude reverts to the domain-general implicit-vs-explicit heuristic, which happens to correlate strongly with \$D\$ on the DR6 snippet set.

Practical consequence for verifier designers: the prompt shape matters non-monotonically. A specific description of \$D\$ works. A domain-general question about implicit assumptions may also work – for targets that align with the implicit-vs-explicit axis. A prompt that sits in between, naming the domain but not the target, may work *worse* than either extreme because it primes surface features that don't discriminate.

6. What DR6g does not license

(naive-UTC) aligns with the implicit-vs-explicit axis by construction. A target that does NOT align – say, "this code assumes exclusive file access" – may not be catchable by domain-blind LLM verifiers, because both correct and incorrect implementations may be equally implicit or explicit about it.

one snippet set with one LLM. The non-monotonic dip may not reproduce for other targets or on other verifiers.

matters – DR6d showed regex hits the wall under open enumeration. DR6g shows the wall can be escaped even under open enumeration for targets with strong domain-general proxies.

- **The wall is easy to escape for all code targets.** DR6g's target
- **The DR6 gradient is universal.** DR6e/f/g measured one target on
- **DR5's condition (a) is now optional.** Open enumeration still

7. Next follow-ups

assumes exclusive file access" or "this code assumes the caller holds a lock." Predicted: fully domain-blind LLM verifier hits the wall because no implicit-vs-explicit correlate exists.

specificity systematically; measure overlap gap; fit a curve.

- **DR6h – a target without a domain-general proxy.** Try "this code
- **DR8 – quantitative gradient study.** Vary the prompt's D-

Neither is required for the main DR5* claim. Both would fill in the graded picture DR6f/g opened.

Appendix: reproduction

```
uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6g
```

Reads three verifier_full_blind_A/B/C.json files, aggregates by median, applies the four G-gates, writes results/dr6g_verdict.json.